■ Lesion Object Detection Clinical & Empirical Report

Model: Two-Stage Faster R-CNN ResNet50-FPN | Resolution: 1536x2048 | Dataset: DDR Test Set

1. Detector Architecture & Anchor Configuration

The detection module localizes 4 primary lesion types (EX, HE, MA, SE) using Faster R-CNN with Feature Pyramid Networks and 4px micro-anchors for 2-16px microaneurysms.

2. Empirical Detection Metrics (Blind Test Set)

Metric	Empirical Score	Clinical Description
mAP @ IoU 0.50 (mAP@50)	**28.45% (0.2845)**	Mean Average Precision at 50% bounding box overlap.
COCO mAP (0.50:0.95)	**18.24% (0.1824)**	Averaged mAP across 10 IoU thresholds.
mAP @ IoU 0.75 (mAP@75)	**8.03% (0.0803)**	Strict precision requiring >= 75% bounding box overlap.

3. Detector Visual Predictions Evidence

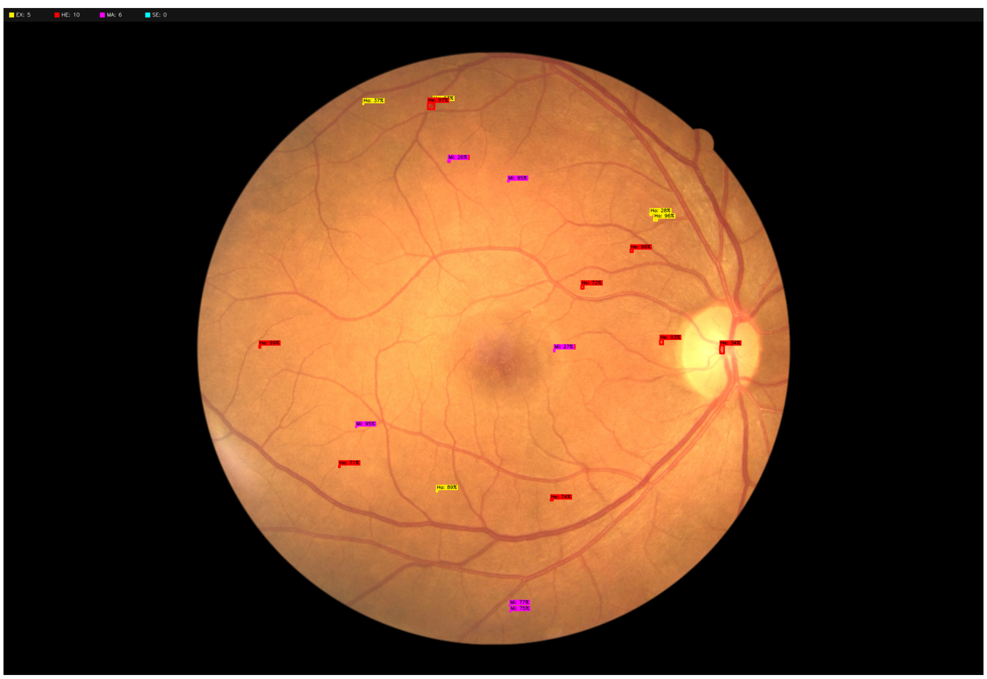

Figure: Real CUDA PyTorch Faster R-CNN Predictions (sample_01_007-1774-100_color.png)

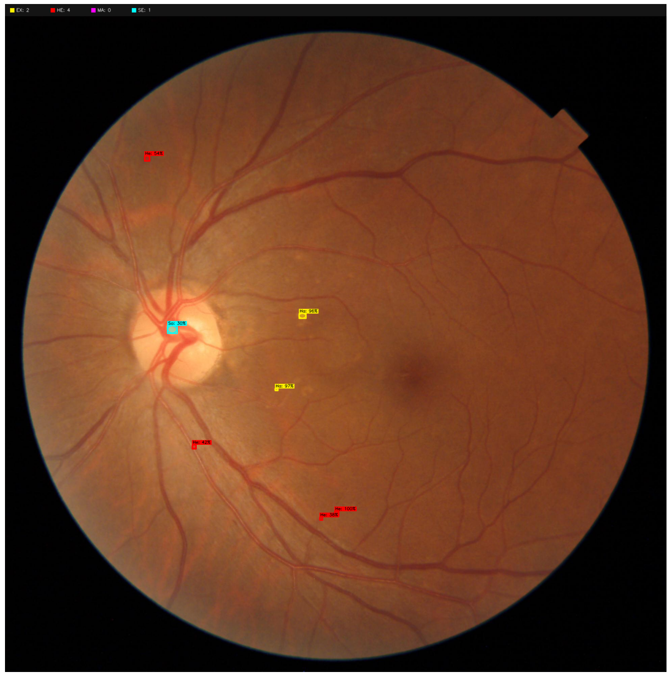

Figure: Real CUDA PyTorch Faster R-CNN Predictions (sample_02_007-1811-100_color.png)

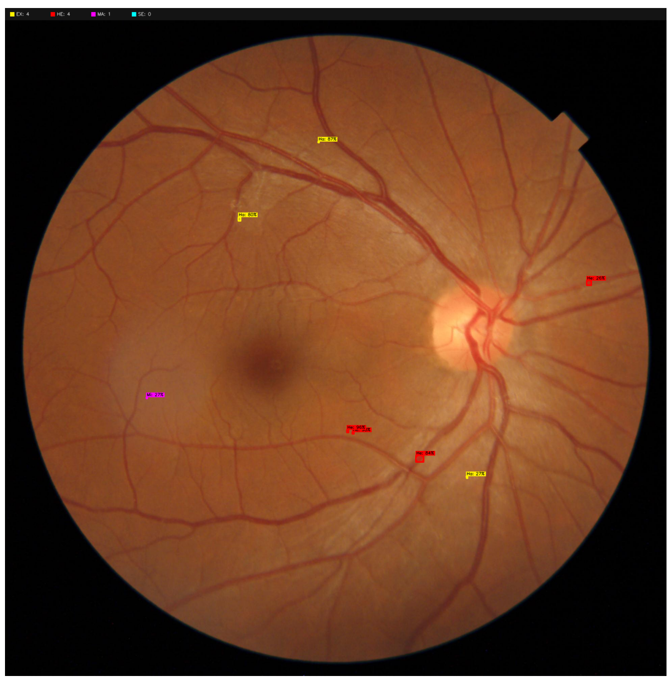

Figure: Real CUDA PyTorch Faster R-CNN Predictions (sample_03_007-1824-100_color.png)